WORKED SOLUTION

STEP III 2017, Question 10

Mechanics • Hinged rod

STEP III 2017, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s3-q10

The key idea

While the particle does not slip, rod and particle turn together as one rigid body about the hinge, so conservation of energy gives $\dot\theta$ straight away, and differentiating gives $\ddot\theta$. The particle then moves on a circle about P, so its acceleration has the familiar radial and transverse components. Newton's second law for the particle alone, resolved along and perpendicular to the rod, gives the friction and normal reaction, and slipping starts when friction reaches μ times the reaction.

Solution

The energy equation

Model the particle as a point mass and the hinge as smooth. The moment of inertia of the uniform rod about its end P is $\frac{1}{3}m(3a)^2 = 3ma^2$, and the particle, a distance ℓ from P, contributes $m\ell^2$. While there is no slipping, the system is a rigid body turning about P with angular speed $\dot\theta$, and its kinetic energy is

$$\tfrac{1}{2}\left(3ma^2 + m\ell^2\right)\dot\theta^2.$$

When the rod has turned through θ, its centre (at distance $\frac{3}{2}a$ from P) has fallen $\frac{3}{2}a\sin\theta$ and the particle has fallen $\ell\sin\theta$. The hinge reaction does no work (its point of application is fixed), and friction and the normal reaction are internal to the system and do no net work when there is no slipping. Starting from rest, conservation of energy gives

$$\tfrac{1}{2}m\left(3a^2 + \ell^2\right)\dot\theta^2 = mg\left(\tfrac{3}{2}a + \ell\right)\sin\theta,$$

which, multiplied by $2/m$, is

$$(3a^2 + \ell^2)\dot\theta^2 = g(3a + 2\ell)\sin\theta,$$

as required.

The angular acceleration

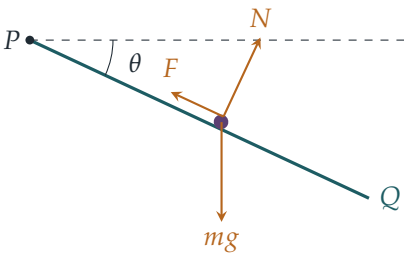
Differentiating with respect to t,

$$2(3a^2+\ell^2)\dot\theta\,\ddot\theta = g(3a+2\ell)\cos\theta\,\dot\theta.$$

For $t>0$ we have $\dot\theta\neq 0$, so we may divide by $2\dot\theta$ (the result at $t=0$ follows by continuity, or directly by taking moments about P):

$$\ddot\theta = \frac{g(3a+2\ell)\cos\theta}{2(3a^2+\ell^2)}.$$

Forces on the particle



The rod lies at angle θ below the horizontal. The rod pushes on the particle with a normal reaction N perpendicular to the rod (away from it, since the particle sits on top) and a friction force F along the rod, which we take to act towards P. The particle moves on a circle of radius ℓ about P, so its acceleration has a component $\ell\ddot\theta$ perpendicular to the rod in the direction of increasing θ (down and away from the rod's upper face), and a component $\ell\dot\theta^2$ along the rod towards P.

Perpendicular to the rod, in the direction of increasing θ, the weight has component $mg\cos\theta$ and N acts the other way:

$$mg\cos\theta - N = m\ell\ddot\theta.$$

Along the rod, towards P, the weight has component $-mg\sin\theta$ (it pulls the particle away from P):

$$F - mg\sin\theta = m\ell\dot\theta^2.$$

The normal reaction

Using $\ddot\theta$ from above,

$$\begin{aligned} N &= mg\cos\theta - m\ell\ddot\theta = mg\cos\theta\left(1-\frac{\ell(3a+2\ell)}{2(3a^2+\ell^2)}\right)\\ &= mg\cos\theta\,\frac{6a^2+2\ell^2-3a\ell-2\ell^2}{2(3a^2+\ell^2)} = \frac{3mga(2a-\ell)\cos\theta}{2(3a^2+\ell^2)}. \end{aligned}$$

Since $\ell<2a$, the factor $2a-\ell$ is positive, and $\cos\theta>0$ for acute θ. So $N>0$; in particular the normal reaction is non-zero, and the particle stays in contact with the rod, while θ is acute.

The point of slipping

From the second equation and the energy equation,

$$F = mg\sin\theta + m\ell\dot{\theta}^2 = mg\sin\theta\left(1 + \frac{\ell(3a+2\ell)}{3a^2+\ell^2}\right)$$
$$= mg\sin\theta\,\frac{3a^2+3a\ell+3\ell^2}{3a^2+\ell^2} = \frac{3mg(a^2+a\ell+\ell^2)\sin\theta}{3a^2+\ell^2}.$$

This is positive for $0 < \theta < \frac{1}{2}\pi$, so friction does act towards P, as assumed: without it the particle would slide outwards along the rod.

Ask yourself Why is the condition for slipping $F = \mu N$, rather than an inequality?

While the particle grips the rod, friction can supply any force up to μN, so we need $F \le \mu N$. At $\theta = 0$ we have $F = 0 < \mu N$. As θ increases, F grows (like $\sin\theta$) and N shrinks (like $\cos\theta$), so F/N, which is a positive multiple of $\tan\theta$, increases steadily. The particle is on the point of slipping at the first angle where $F = \mu N$:

$$\frac{3mg(a^2+a\ell+\ell^2)\sin\theta}{3a^2+\ell^2} = \mu\,\frac{3mga(2a-\ell)\cos\theta}{2(3a^2+\ell^2)}.$$

Cancelling $3mg/(3a^2+\ell^2)$ and dividing by $\cos\theta \neq 0$,

$$\tan\theta = \frac{\mu a(2a-\ell)}{2(\ell^2+a\ell+a^2)},$$

as required.

If $\ell > 2a$

The formula for N was derived assuming the particle moves with the rod. If $\ell > 2a$, it gives $N < 0$ at $\theta = 0$, which is impossible: the rod can only push on the particle, not pull it. So the particle cannot move with the rod, even at the instant of release.

To see what does happen, consider the rod on its own. Taking moments about P at the moment of release,

$$3ma^2\,\ddot{\theta} = mg\cdot\tfrac{3}{2}a \quad\Longrightarrow\quad \ddot{\theta} = \frac{g}{2a},$$

so the point of the rod beneath the particle starts with downward acceleration $\ell\ddot{\theta} = \frac{\ell}{2a}g$, which is greater than g when $\ell > 2a$. The rod falls away faster than the particle can fall freely. So at the moment of release the particle leaves the rod (the reaction between them is zero) and begins to fall freely under gravity, while the rod swings down on its own.

Checks

- Taking moments about P for the whole system gives $(3ma^2+m\ell^2)\ddot{\theta} = mg\cos\theta\,(\frac{3}{2}a+\ell)$, the same $\ddot{\theta}$ as differentiating the energy equation.

- The boundary case $\ell = 2a$ fits both pictures: $N = 0$ at release, and the rod's point at distance $2a$ starts with acceleration exactly g.
- With $a = \ell = 1$ and $\mu = 1$, the slipping angle has $\tan\theta = \frac{1}{6}$, and at that angle the formulas above give $F/N = 6\tan\theta = 1 = \mu$. The formulas for N and F were also confirmed by computer, by stepping the motion numerically and computing the particle's acceleration directly from its position.

Common slips

- Using the moment of inertia about the centre, $\frac{1}{12}m(3a)^2$, instead of about the hinge; the moment of inertia of a rod of length L about an end is $\frac{1}{3}mL^2$.
- Forgetting the centripetal term $m\ell\dot{\theta}^2$ in the friction equation; it is what makes the numerator $a^2 + a\ell + \ell^2$ rather than a^2.
- Getting the sign of $\ell\ddot{\theta}$ wrong in the normal equation: the rod is accelerating away from the particle, so N is less than $mg\cos\theta$, not more.

Beyond STEP

The last part is a classic surprise: a point on a falling hinged rod can accelerate faster than g, which is why a coin placed near the end of a falling ruler (or a chimney toppling over) gets left behind. The systematic way to find constraint forces like N and F is to write the equations of motion in polar coordinates, and Lagrangian mechanics later shows how to get the motion without them at all. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body rotation and these polar-coordinate accelerations carefully.

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